THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** 20	Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$.
2	**(b)** -14	Put $p(z) = 0$.
3	**(a)** $\frac{39}{65}$	Reduce the fraction; only 2 and 5 may remain in the denominator.
4	**(d)** $49x^2 + 56xy + 16y^2$	Use the square identity.
5	**(d)** 6	Chord $= 2\sqrt{r^2 - d^2}$.
6	**(c)** 22	Arc $= \frac{\theta}{360} \times 2\pi r$.
7	**(a)** $\frac{1}{6}$	There are 36 equally likely outcomes.
8	**(a)** 4	$d = \frac{t_n - t_1}{n-1} = \frac{40}{10}$.
9	**(c)** 9 is divisible by 3 but not by 6	Counterexample for the converse.
10	**(a)** 87	$\frac{12 \times 89 + 63}{13} = 87$.
11	**(d)** 9	Pairs: $(360, 156) \to (204, 156) \to (156, 48) \to (108, 48) \to (60, 48) \to (48, 12) \to (36, 12) \to (24, 12) \to (12, 12) \to (12, 0)$.
12	**(c)** 150°	One part $= 360^\circ \div 12 = 30^\circ$.
13	**(c)** -1	Rewrite as $y = -x + \ldots$; slope $= -\frac{a}{b}$.
14	**(d)** 36	Surface area $\propto r^2$, so $6^2 = 36$.
15	**(b)** $(4.5,\ 4.5)$	Average the coordinates.
16	**(a)** 3	Slope $= \frac{y_2 - y_1}{x_2 - x_1}$.
17	**(b)** 4	$400 = 2^4 \times 5^2$
18	**(b)** $6x^2 + 19x + 15$	Multiply term by term.
19	**(d)** 5	$r^2 = h^2 + d^2$.
20	**(a)** 12	$h = \frac{2A}{b}$.
21	**(a)** $\frac{1}{6}$	Count favourable outcomes out of 36.
22	**(d)** 9	$a = t_n - (n-1)d = 69 - 60$.
23	**(b)** 6	6 is even.
24	**(b)** 85	$\frac{27 \times 84 - 58}{26} = 85$.
25	**(b)** 9	Pairs: $(420, 364) \to (364, 56) \to (308, 56) \to (252, 56) \to (196, 56) \to (140, 56) \to (84, 56) \to (56, 28) \to (28, 28) \to (28, 0)$.
26	**(c)** 55°	Let angles be x and $x + d$; they add to 180°.
27	**(a)** 14	Equal slopes between consecutive points.
28	**(a)** 125	Volume $\propto a^3$, so $5^3 = 125$.
29	**(d)** $(-1, -3)$	The centre is the midpoint of a diameter.
30	**(b)** -3	Put $y = 0$.
31	**(a)** $\frac{4}{33}$	$x = \frac{ab}{99}$, then reduce.
32	**(b)** 45	$a^2 + b^2 = (a - b)^2 + 2ab$.
33	**(c)** 58°	Isosceles triangle OAB.
34	**(a)** 12	$h = \frac{A}{b}$.
35	**(b)** $\frac{2}{3}$	Count the outcomes in either event, without double counting.
36	**(c)** 59	$5d = 20$ so $d = 4$; then $t_{15} = t_9 + 6d$.
37	**(b)** 4	$n = 4$ gives 121, which is composite.
38	**(a)** 64	Sum of the extra numbers $= 70 \times 60 - 40 \times 73 = 1280$; divide by 20.
39	**(d)** 23	Multiples of 3: 16; of 5: 10; of both: 3.

40	**(c)** 480 cm^2	Other side $= 30$ cm; area $= 16 \times 30$.
41	**(d)** 8	Equal slopes: $-\frac{4}{2} = -\frac{k}{4}$, so $k = 8$.
42	**(d)** $1 : 216$	Ratio of volumes $= r_1^3 : r_2^3 = 1 : 216$.
43	**(c)** $(3, 0)$	Set $PA = PB$ with $P = (x, 0)$.
44	**(b)** 4	Substitute y and solve for x.
45	**(c)** 6	Find how many digits repeat.
46	**(d)** 7	Square both sides.
47	**(c)** 72°	$\angle A + \angle C = 180^\circ$.
48	**(c)** 52	Side $= \sqrt{A}$.
49	**(d)** $\frac{7}{12}$	Both favourable and total counts increase.
50	**(b)** 182	$2 + 4 + \cdots + 2n = n(n+1)$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** 12	List the factors of 72 in pairs; the count is 12 (even, so 72 is not a perfect square).
2	**(a)** 24	$12(73-71)=n_2(74-73)$.
3	**(a)** 18	The algorithm adds the digits.
4	**(d)** 52 cm	Side $=13$ cm, perimeter $=4\times13$.
5	**(c)** 3	Intercepts 2 and 3; area $=\frac{1}{2}\times2\times3$.
6	**(d)** 924 cm^2	$\pi r(l+r)$ with the given values.
7	**(c)** Quadrant III	Use the sign conditions on the coordinates.
8	**(b)** $8x-5$	Add like terms.
9	**(d)** 6	Multiply under the root.
10	**(c)** 1006009	Write $1003=1000+3$
11	**(d)** 48°	Central angle is double.
12	**(c)** 140	Area $=\frac{1}{2}d(h_1+h_2)$.
13	**(a)** $\frac{4}{11}$	Count the matching letters, including repeats.
14	**(d)** 196	$1+3+5+\cdots+(2n-1)=n^2$.
15	**(d)** right-angled	$33^2+56^2=4225$ and $65^2=4225$.
16	**(c)** 18%	$\dfrac{360\times35+680\times9}{1040}=18\%$.
17	**(d)** 100	LCM $=\dfrac{20\times25}{5}$.
18	**(a)** 120 cm^2	Area $=\frac{1}{2}d_1d_2$.
19	**(d)** 259	Rate $=\dfrac{110}{10}=11$ per km; fixed charge 28.
20	**(c)** 17 cm	$l=\sqrt{8^2+15^2}$.
21	**(c)** 36	Add the three side lengths.
22	**(a)** 4	$p(2)-p(1)=a$.
23	**(c)** rational	Collect the rational and irrational parts.
24	**(d)** 5600	Use $a^2-b^2=(a+b)(a-b)$.
25	**(a)** 135°	Opposite angles of the cyclic quadrilateral are supplementary.
26	**(c)** 126	Find the breadth first.
27	**(d)** 7200	Expected number = total × probability.
28	**(d)** 3	$t_5/t_3=r^2$.
29	**(b)** $n(n+1)$ is even	One of two consecutive integers is even.
30	**(b)** 74	$\dfrac{62\times3+78\times4+79\times4}{11}=74$.
31	**(d)** 19	$\lfloor\sqrt{373}\rfloor=19$.
32	**(d)** 43 cm	Each side of the new figure is half a diagonal; there are two of each.
33	**(d)** 6	$2(-4)+(6)k=28$ gives $k=6$.
34	**(b)** 693000	$V=693$ m^3.
35	**(a)** $(-4,-9)$	Both signs change.
36	**(a)** 82	Substitute n.
37	**(c)** $\frac{3}{7}$	Take the average.
38	**(b)** $(x-7)(x+4)$	Find the pair of numbers.
39	**(d)** 12	Equal chords are equidistant from the centre.

40	**(c)** 11	Perimeter is unchanged.
41	**(b)** $\frac{7}{12}$	Use the complement rule.
42	**(d)** 19	These form an AP; count with $n = \frac{l-a}{d} + 1$.
43	**(b)** 95	It equals 5×19 (number of terms times the middle term).
44	**(b)** 70	New salary is higher by $n \times 3 = 42$ thousand.
45	**(a)** 9	Pairs: $(312, 200) \to (200, 112) \to (112, 88) \to (88, 24) \to (64, 24) \to (40, 24) \to (24, 16) \to (16, 8) \to (8, 8) \to (8, 0)$.
46	**(a)** 44	Opposite angles are equal.
47	**(c)** $(-1, 7)$	$8(-1) + 5(7) = 27$
48	**(b)** 72	$\frac{30}{5} \times \frac{15}{5} \times \frac{20}{5}$.
49	**(b)** 15	Area $=$ length $\times$ breadth.
50	**(c)** parallel	Use the slope-intercept form.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** 11	Apply Brahmagupta's rules in order.
2	**(c)** 32	$4xy = (x+y)^2 - (x-y)^2$.
3	**(c)** 66°	Right angle at C.
4	**(c)** 110	Distance $= n\times$ circumference.
5	**(a)** $\frac{1}{2}$	Use the 8 outcomes.
6	**(a)** 70	$t_{20} - t_{10} = 10d$.
7	**(d)** 7	Only perfect squares have an odd number of factors.
8	**(d)** 15%	$\frac{3000\times 9 + 2000\times 24}{5000} = 15\%$.
9	**(b)** 3	Pairs: $(453,384)\to(384,69)\to(315,69)\to(246,69)\to(177,69)\to(108,69)\to(69,39)\to(39,30)\to(30,9)\to(21,9)\to(12,9)\to(9,3)\to(6,3)\to(3,3)\to(3,0)$.
10	**(d)** 60 cm	Midpoint theorem: $BC = 2DE$.
11	**(d)** 7	$y = mx + c$ with $y = 7$ at $x = 0$: $c = 7 - (6)(0)$.
12	**(a)** 1152 cm^3	$\frac{1}{3} \times$ base area $\times$ height.
13	**(d)** 65	Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$.
14	**(b)** -30	Put $p(z) = 0$.
15	**(d)** $\frac{21}{70}$	Reduce the fraction; only 2 and 5 may remain in the denominator.
16	**(b)** $36x^2 - 60xy + 25y^2$	Use the square identity.
17	**(b)** 12	Chord $= 2\sqrt{r^2 - d^2}$.
18	**(a)** 88	Arc $= \frac{\theta}{360} \times 2\pi r$.
19	**(a)** $\frac{1}{9}$	There are 36 equally likely outcomes.
20	**(a)** -3	$d = \frac{t_n - t_1}{n-1} = \frac{-24}{8}$.
21	**(d)** a 4 cm by 6 cm rectangle is not a square	Counterexample for the converse.
22	**(b)** 85	$\frac{37\times 86 + 48}{38} = 85$.
23	**(a)** 6	Pairs: $(294,245)\to(245,49)\to(196,49)\to(147,49)\to(98,49)\to(49,49)\to(49,0)$.
24	**(a)** 144°	One part $= 360^\circ \div 15 = 24^\circ$.
25	**(c)** $-\frac{1}{2}$	Rewrite as $y = -\frac{1}{2}x + \ldots$; slope $= -\frac{a}{b}$.
26	**(a)** 16	Surface area $\propto r^2$, so $4^2 = 16$.
27	**(d)** $(-1.5,\ 6.5)$	Average the coordinates.
28	**(a)** 3	Slope $= \frac{y_2 - y_1}{x_2 - x_1}$.
29	**(a)** 3	$250 = 2^1 \times 5^3$
30	**(c)** $6x^2 + 25x + 14$	Multiply term by term.
31	**(b)** 10	$r^2 = h^2 + d^2$.
32	**(a)** 8	$h = \frac{2A}{b}$.
33	**(d)** $\frac{1}{6}$	Count favourable outcomes out of 36.
34	**(d)** 30	$a = t_n - (n-1)d = 16 + 14$.
35	**(c)** 12	12 is divisible by 4 and 6 but not by 24.
36	**(a)** 63	$\frac{31\times 66 - 156}{30} = 63$.
37	**(c)** 9	Pairs: $(448,176)\to(272,176)\to(176,96)\to(96,80)\to(80,16)\to(64,16)\to(48,16)\to(32,16)\to(16,16)\to(16,0)$.

38	**(b)** 50°	Let angles be x and $x + d$; they add to 180°.
39	**(a)** -7	Equal slopes between consecutive points.
40	**(b)** 216	Volume $\propto a^3$, so $6^3 = 216$.
41	**(c)** $(3, 0)$	The centre is the midpoint of a diameter.
42	**(c)** 2	Put $y = 0$.
43	**(b)** $\frac{6}{11}$	$x = \frac{ab}{99}$, then reduce.
44	**(a)** 73	$a^2 + b^2 = (a - b)^2 + 2ab$.
45	**(a)** 48°	Isosceles triangle OAB.
46	**(d)** 8	$h = \frac{A}{b}$.
47	**(c)** $\frac{2}{3}$	Count the outcomes in either event, without double counting.
48	**(d)** -8	$5d = -10$ so $d = -2$; then $t_{15} = t_9 + 6d$.
49	**(d)** 3	$n = 3$ gives 119, which is composite.
50	**(b)** 115	Sum of the extra numbers $= 65 \times 57 - 38 \times 40 = 2185$; divide by 19.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND